

# Manas Dubey

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## EDUCATION

### Manipal University Jaipur

*B.Tech (Hons.) Computer Science Engineering (AI & ML)*

Rajasthan, IN

*Aug. 2022 – June 2026*

### The Air Force School

*CBSE Board Examination (X & XII)*

New Delhi, IN

*Apr. 2020 – May 2022*

## EXPERIENCE

### NeoFab.AI

Seattle, WA

*Software Engineer Intern (Full-Stack + AI)*

*Aug. 2025 – Present*

- Engineered a Python ETL pipeline (Firecrawl, regex parsing) crawling manufacturer pages into Firestore, scaling the dataset  $\sim 17\times$  (54 to 962 records) across 4 sectors.
- Developed the Next.js + TypeScript landing page with a live Three.js factory simulation, used as the company's investor deck, plus a config system (workflow, process-flow, sidebar editors) on FastAPI endpoints with RBAC.
- Designed a procurement agent that auto-sends requirement-based RFQs to scraped vendor emails and threads replies into an in-dashboard inbox for direct supplier chat.
- Grew a RAG knowledge base to 908 machine records across 4 manufacturing domains; shipped to a live Industrial AI product evaluated by OLA, Ather, and Siemens.

## PROJECTS

### Legally AI | *Next.js, FastAPI, TypeScript, FAISS, InLegalBERT, Docker*

[Live](#) | [Code](#)

- Architected an AI legal search tool over 88,988 passages from 1,991 Supreme Court judgments, using a multi-stage RAG pipeline (query reformulation, vector search, citation verification) to cut hallucinated citations.
- Shipped a real-time multilingual (Hindi/English/Hinglish) voice assistant (WebRTC + Deepgram STT  $\rightarrow$  Groq Llama 3.3 70B  $\rightarrow$  TTS) at  $<800$  ms P50 latency with interruptible turn-taking and cited summaries.
- Added a reasoning layer combining precedent retrieval, InLegalBERT classification, and grounded LLM inference, returning confidence scores with source-linked evidence.
- Implemented a PDF/DOCX/OCR ingestion pipeline for clause extraction, document chat, and statute identification.

### Conduit | *Python, FastAPI, Redis, PostgreSQL, Docker, MCP, Prometheus, Grafana*

[Code](#)

- Architected a drop-in OpenAI-compatible LLM gateway over 5 providers; cut API cost 80% on a 2,000-request load test via semantic caching and complexity-based routing.
- Engineered a Redis-backed semantic cache (vector similarity, TTL tiers) hitting 70% cache rate and 77 ms P95 versus 9.1 s, a 99% latency cut.
- Built a fault-tolerant resilience layer (token-bucket rate limiting, budget caps, exponential backoff, circuit breakers, cross-provider failover) with zero caller-visible downtime.
- Shipped an MCP server (`conduit-mcp`) letting any MCP client monitor and control the gateway in natural language over the admin API, plus full observability (Prometheus, Grafana) and a CI/CD pipeline.

### Portfolio | *Next.js, TypeScript, Tailwind CSS, Framer Motion, Vercel*

[Live](#) | [Code](#)

- Built and deployed a personal portfolio (Next.js, TypeScript, Tailwind CSS, Framer Motion) on Vercel with a responsive, animated, accessible UI.
- Added per-project case-study pages (e.g., Legally AI) with write-ups, live demos, source links, and SEO-optimized static rendering.

## TECHNICAL SKILLS

**Languages:** Python, TypeScript, JavaScript, SQL, C++, Java

**Frameworks & Web:** React, Next.js, FastAPI, Node.js, Express.js, Tailwind CSS

**AI/ML:** PyTorch, Hugging Face Transformers, FAISS, pgvector, RAG, Vector Search, InLegalBERT, NLP

**Data & Backend:** AWS, ETL Pipelines, REST APIs, PostgreSQL, Redis, Cloud Firestore, MongoDB, Web Scraping, RBAC, NoSql

**DevOps & Tools:** Git, Docker, CI/CD, Google Cloud, Firebase, Prometheus, Grafana, pytest, Postman, Rest-APIs

## ACHIEVEMENTS

- Received the Student Excellence Award from the Dept. of AIML, MUJ, and Dean's List Award.
- Won Rs.12,00,000+ in seed funding from MUJ E-Cell for Legally AI, an affordable legal-research tool for Indian users.